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Student Number

2016

Mathematics Extension 2

HSC Course Assessment Task 3

2nd June 2016

General Instructions

- Reading time – 5 minutes
- Working time 1 hour
- Write using blue or black pen
Black pen is preferred
- Approved calculators may be used
- In Questions 6 to 8, show relevant mathematical reasoning and/or calculations
- Answer each question in a separate writing booklet
- This paper must not be removed from the examination room
- A reference sheet is provided at the back of this paper
- Diagrams are NOT to scale

Total Marks – 42

Section I - Pages 2 - 3

5 marks

- Attempt Questions 1 to 5
- Allow about 7 minutes for this section

Section II - Pages 4 - 7

37 marks

- Attempt Questions 6 to 8
- Allow about 53 minutes for this section

Question	Marks
1-5	/5
6	/15
7	/14
8	/8
Total	/42

THIS QUESTION PAPER MUST NOT BE REMOVED FROM THE EXAMINATION ROOM

This assessment task constitutes 20% of the HSC Course Assessment

Section I

5 marks

Attempt Questions 1 to 5

Allow about 7 minutes for this section

Use the multiple-choice answer sheet for questions 1-5 (Detach from paper)

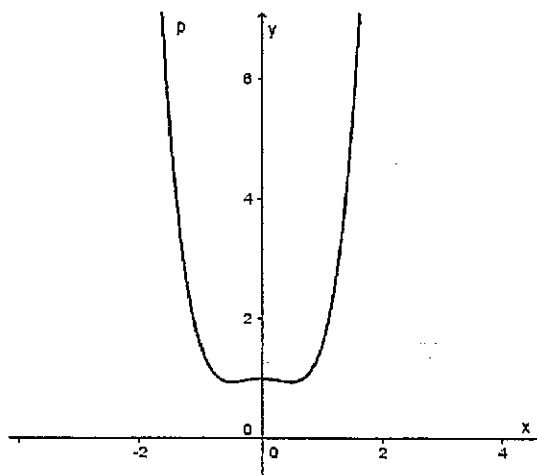
- 1 A suitable substitution to evaluate $\int \sqrt{9 - x^2} \, dx$ is
- (A) $x = 3\sin\theta$
 - (B) $x = 3\tan\theta$
 - (C) $x = 3\operatorname{cosec}\theta$
 - (D) $x = 3\sec\theta$
- 2 $\int \frac{x^2+6}{x^2+4} \, dx$ is equal to
- (A) $x + \frac{1}{2}\ln(x^2 + 4) + c$
 - (B) $x + \ln(x^2 + 4) + c$
 - (C) $x + \frac{1}{2}\tan^{-1}\frac{x}{2} + c$
 - (D) $x + \tan^{-1}\frac{x}{2} + c$
- 3 Using the substitution $t = \tan\frac{\theta}{2}$, $\int \frac{dx}{1-\cos x}$ is equal to
- (A) $\int \frac{2}{t^2} \, dt$
 - (B) $\int \frac{1+t^2}{t^2} \, dt$
 - (C) $\int \frac{1}{t^2} \, dt$
 - (D) $\int \frac{2(1+t^2)}{t^2} \, dt$

- 4 The region bounded by the curve $y = x - \frac{x^2}{12}$ above the x-axis, to the right of the line $x = 6$, is rotated about the line $x = 6$. The integral evaluated to obtain this volume is:

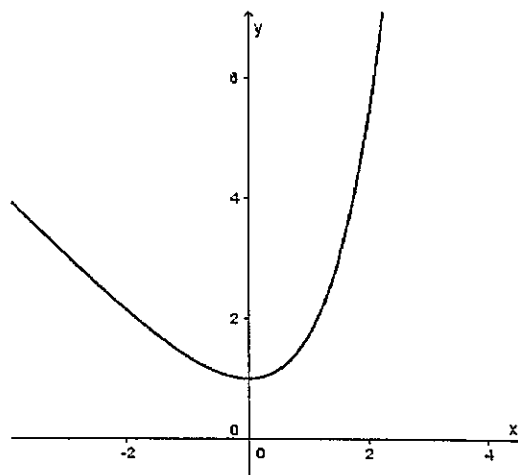
- (A) $V = \pi \int_0^6 (y - 6)^2 dy$
 (B) $V = \pi \int_6^{12} (36 - 12y) dy$
 (C) $V = \pi \int_0^3 (y - 6)^2 dy$
 (D) $V = \pi \int_0^3 (36 - 12y) dy$

- 5 Which curve is the graph of $y = e^x - x$?

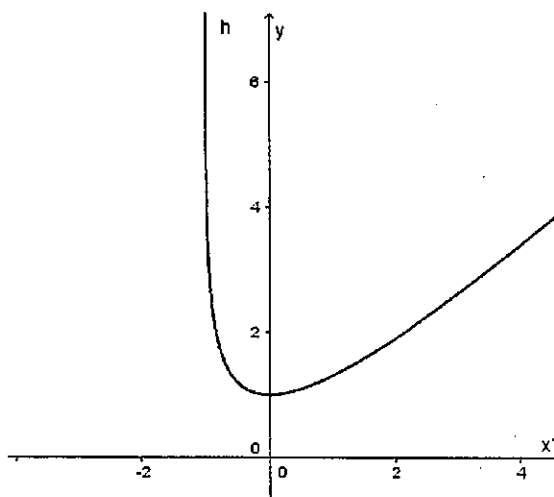
(A)



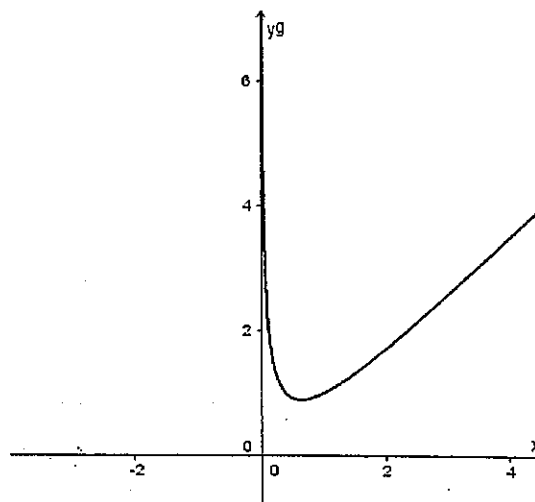
(B)



(C)



(D)



Section II

37 marks

Attempt Questions 6 to 8

Allow about 53 minutes for this section

Answer each question in a separate writing booklet. Extra writing booklets are available.

In Questions 6 to 8, your responses should include relevant mathematical reasoning and/or calculations.

Question 6 (15 marks) Use a SEPARATE writing booklet

(a) Find $\int \frac{x}{\sqrt{9x^2-16}} dx$ 2

(b) Find $\int_4^5 \frac{x^2+5x-20}{x-3} dx$ 3

(c) Find $\int \sin^5 \theta d\theta$ 3

(d) (i) Find real numbers A, B and C such that 2

$$\frac{4}{x^2(x-2)} = \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x-2}$$

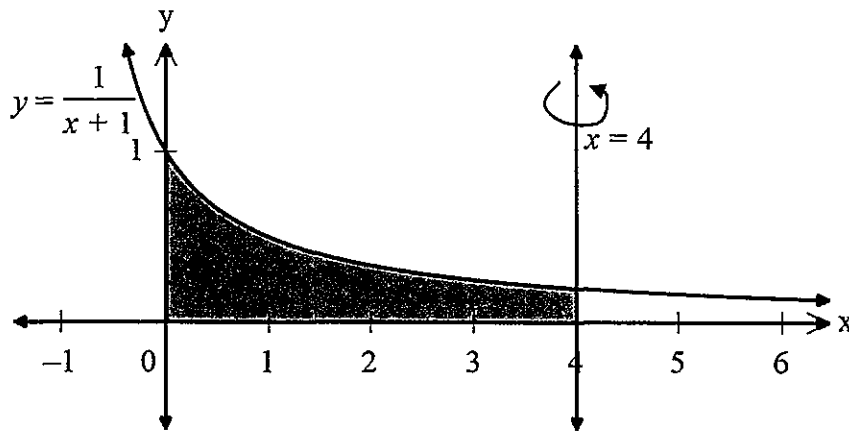
(ii) Hence, find $\int \frac{4}{x^2(x-2)} dx$ 2

(e) Find $\int \cos(\ln x) dx$ 3

End of Question 6

Question 7 (14 marks) Use a SEPARATE writing booklet

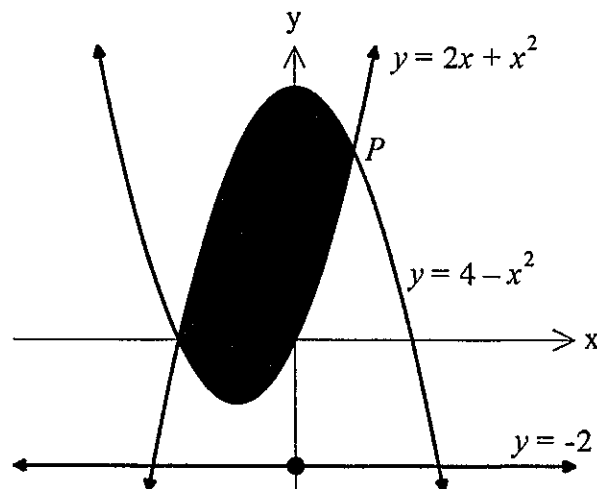
- (a) The diagram shows the graph of $y = \frac{1}{x+1}$. The shaded region between the curve and the x-axis for $0 \leq x \leq 4$ is rotated about the line $x = 4$ to form a solid.



Use the method of cylindrical shells to find the volume of the solid.

4

- (b)



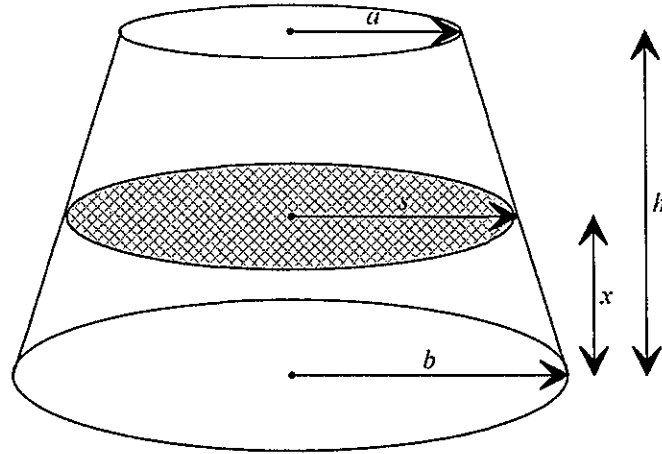
The shaded region bounded by $y = 4 - x^2$ and $y = 2x + x^2$ is rotated about the line $y = -2$. The point P is the intersection of $y = 4 - x^2$ and $y = 2x + x^2$ in the first quadrant.

- | | | |
|-------|--|---|
| (i) | Find the x co-ordinate of P. | 1 |
| (ii) | Use the method of slicing to express the volume of the resulting solid of revolution as an integral. | 3 |
| (iii) | Evaluate the integral in part (ii). | 2 |

Question 7 continues on page 6

Question 7 (continued)

- (c) The diagram shows the frustum of a right cone. (A frustum of a cone is a cone with its top cut off.) The height of the frustum is h metres. Its base is a circle of radius b metres, and its top is a circle of radius a metres. (with $b > a > 0$)



A horizontal cross-section of the frustum, taken at height x metres, is a circle of radius s metres, shown shaded in the diagram.

- (i) Show that $s = b - \frac{(b-a)}{h} x$. 2
- (ii) By integration, find the volume of the frustum. 2

End of Question 7

Question 8 (8 marks) Use a SEPARATE writing booklet

(a) A curve is defined implicitly by $x^3 + y^3 = -1$.

(i) Use implicit differentiation to find $\frac{dy}{dx}$. 2

(ii) Find the equations of the tangents to the curve $x^3 + y^3 = -1$ at the x and y intercepts and at the point where the curve meets the line $y = x$. 2

(iii) Show that $x^3 + y^3 = -1$ can be rewritten as $y = -x \left(\sqrt[3]{1 + \frac{1}{x^3}} \right)$ and 2
hence find the equation of the asymptote to the curve $x^3 + y^3 = -1$.

(iv) Sketch the curve $x^3 + y^3 = -1$. 2

End of Question 8

End of Examination

2016 Mathematics Extension 2
HSC Assessment Task 3
2nd June 2016
Sample Solutions.

Section 1 - Multiple Choice

$$\begin{aligned}
 1.) \int \sqrt{9-x^2} \, dx & \quad \left| \begin{array}{l} \text{let } x = 3 \sin \theta \\ -\frac{\pi}{2} \leq x \leq \frac{\pi}{2} \\ dx = 3 \cos \theta \, d\theta \end{array} \right. \\
 = \int \sqrt{9-9\sin^2 \theta} \times 3 \cos \theta \, d\theta \\
 = \int 3 \cos \theta \times 3 \cos \theta \, d\theta \\
 = 9 \int \cos^2 \theta \, d\theta \\
 \text{etc.}
 \end{aligned}$$

Ⓐ

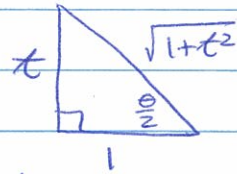
$$\begin{aligned}
 2.) \int \frac{x^2+6}{x^2+4} \, dx \\
 = \int \frac{x^2+4+2}{x^2+4} \, dx \\
 = \int \left(1 + \frac{2}{x^2+4} \right) dx \\
 = x + 2 \times \frac{1}{2} \tan^{-1} \frac{x}{2} + C \\
 = x + \tan^{-1} \frac{x}{2} + C
 \end{aligned}$$

ⓓ

(2)

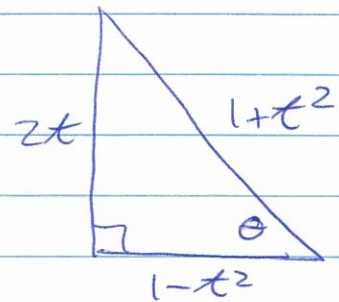
$$\begin{aligned}
 3.) \int \frac{d\theta}{1 - \cos\theta} \\
 &= \int \frac{1}{1 - \frac{1-t^2}{1+t^2}} \times \frac{2}{1+t^2} dt \\
 &= \int \frac{1+t^2}{1+t^2 - (1-t^2)} \times \frac{2}{1+t^2} dt \\
 &= \int \frac{2}{2t^2} dt \\
 &= \int \frac{1}{t^2} dt
 \end{aligned}$$

Let $t = \tan \frac{\theta}{2}$



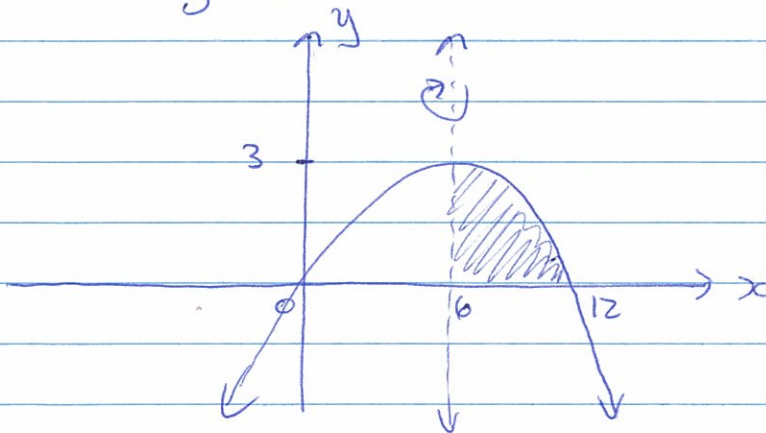
$$\begin{aligned}
 \frac{dt}{d\theta} &= \frac{1}{2} \sec^2 \frac{\theta}{2} \\
 &= \frac{1}{2} (\sqrt{1+t^2})^2 \\
 &= \frac{1+t^2}{2}
 \end{aligned}$$

$$\begin{aligned}
 \frac{d\theta}{dt} &= \frac{2}{1+t^2} \\
 d\theta &= \frac{2}{1+t^2} dt
 \end{aligned}$$



(C)

$$\begin{aligned}
 4.) y &= x \left(1 - \frac{x}{12}\right) \\
 \text{when } y &= 0, x = 0, x = 12.
 \end{aligned}$$



when $x=6, y=3$

$$\begin{aligned}
 \delta V &= \pi r^2 \delta y \\
 &= \pi (x-6)^2 \delta y \\
 &= \pi (x^2 - 12x + 36) \delta y \\
 V &= \pi \int_0^3 (x^2 - 12x + 36) dy \\
 &= \pi \int_0^3 (-12y + 36) dy \\
 &= \pi \int_0^3 (36 - 12y) dy
 \end{aligned}$$

(D)

③

5.) As $x \rightarrow -\infty$, $y = e^x - x \rightarrow y = -x$

$\therefore y = -x$ is an asymptote as $x \rightarrow -\infty$

⑥

6.) a.) $\int \frac{x}{\sqrt{9x^2-16}} dx$

$$= \int x (9x^2-16)^{-\frac{1}{2}} dx$$

$$= 2(9x^2-16)^{\frac{1}{2}} \times \frac{1}{18} + C$$

$$= \frac{1}{9} (9x^2-16)^{\frac{1}{2}} + C$$

$$= \frac{1}{9} \sqrt{9x^2-16} + C$$

b.) $\int_4^5 \frac{x^2+5x-20}{x-3} dx$

$$= \int_4^5 \left(x+8 + \frac{4}{x-3} \right) dx$$

$$= \left[\frac{1}{2}x^2 + 8x + 4 \ln(x-3) \right]_4^5$$

$$= \frac{25}{2} + 40 + 4 \ln 2 - (8 + 32 + 4 \ln 1)$$

$$= \frac{25}{2} + 4 \ln 2$$

$$\begin{array}{r} x+8 \\ x-3 \overline{) x^2+5x-20} \\ \underline{x^2-3x} \\ 8x-20 \\ \underline{8x-24} \\ 4 \end{array}$$

c.) $\int \sin^5 \theta d\theta$

$$= \int \sin^4 \theta \sin \theta d\theta$$

$$= \int (1-\cos^2 \theta)^2 \sin \theta d\theta$$

$$= \int (1-2\cos^2 \theta + \cos^4 \theta) \sin \theta d\theta$$

$$= \int \sin \theta d\theta - 2 \int \cos^2 \theta \sin \theta d\theta + \int \cos^4 \theta \sin \theta d\theta$$

$$= -\cos \theta + \frac{2}{3} \cos^3 \theta - \frac{1}{5} \cos^5 \theta + C$$

(5)

$$6d.) (i) \frac{4}{x^2(x-2)} = \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x-2}$$

$$= \frac{Ax(x-2) + B(x-2) + Cx^2}{x^2(x-2)}$$

$$4 = Ax(x-2) + B(x-2) + Cx^2$$

when $x = 2$,

$$4 = 4C$$

$$C = 1$$

when $x = 0$,

$$4 = -2B$$

$$B = -2$$

coefficient of x^2 ,

$$A + C = 0$$

$$A + 1 = 0$$

$$A = -1$$

$$\therefore A = -1, B = -2, C = 1$$

$$(ii) \int \frac{4}{x^2(x-2)} dx = \int \frac{-1}{x} dx + \int \frac{-2}{x^2} dx + \int \frac{1}{x-2} dx$$

$$= -\ln|x| + 2x^{-1} + \ln|x-2| + C$$

$$= \ln\left|\frac{x-2}{x}\right| + \frac{2}{x} + C \quad (2)$$

- Some left out the absolute signs in 'ln'

- The answer must be simplified as in (2)

(6)

$$6e.) \int \cos(\ln x) dx$$

$$\text{let } u = \cos(\ln x)$$

$$\therefore du = -\sin(\ln x) \times \frac{1}{x} dx$$

$$dv = dx$$

$$\therefore v = x$$

$$I = \int \cos(\ln x) dx$$

$$= x \cos(\ln x) - \int -\sin(\ln x) dx$$

$$= x \cos(\ln x) + \int \sin(\ln x) dx$$

$$\text{let } u = \sin(\ln x)$$

$$du = \frac{1}{x} \cos(\ln x) dx$$

$$dv = dx$$

$$v = x$$

$$I = x \cos(\ln x) + x \sin(\ln x) - \int \cos(\ln x) dx$$

$$= x \cos(\ln x) + x \sin(\ln x) - I$$

$$2I = x \cos(\ln x) + x \sin(\ln x)$$

$$I = \frac{x}{2} \cos(\ln|x|) + \frac{x}{2} \sin(\ln|x|) + C$$

Most students used the substitution $u = e^x$ which was not the most efficient method.

⑦

7a.)

$$\delta V = \pi \left[(r + \delta r)^2 - r^2 \right] h$$

$$= \pi \left[2r\delta r + (\delta r)^2 \right] h$$

$$\delta V = 2\pi r h \delta r \quad \text{since } (\delta r)^2 \text{ is effectively zero.}$$

$$\delta V = 2\pi (4-x) \frac{1}{x+1} \delta x \quad \left| \begin{array}{l} r = 4-x \\ h = y \\ \delta r = \delta x \frac{1}{x+1} \\ y = \frac{1}{x+1} \end{array} \right.$$

$$V = 2\pi \int_0^4 (4-x) \frac{1}{x+1} dx$$

$$= 2\pi \int_0^4 \frac{4-x}{x+1} dx$$

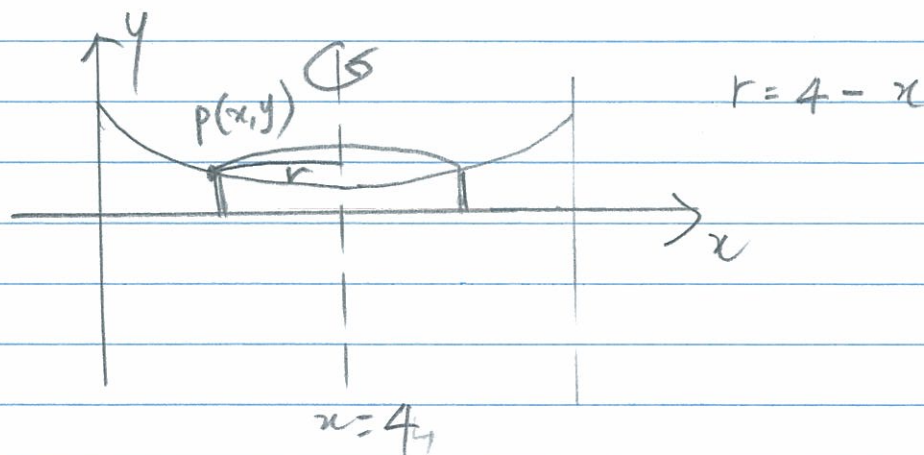
$$= -2\pi \int_0^4 \frac{x+1-5}{x+1} dx$$

$$= -2\pi \int_0^4 \left(1 - \frac{5}{x+1} \right) dx$$

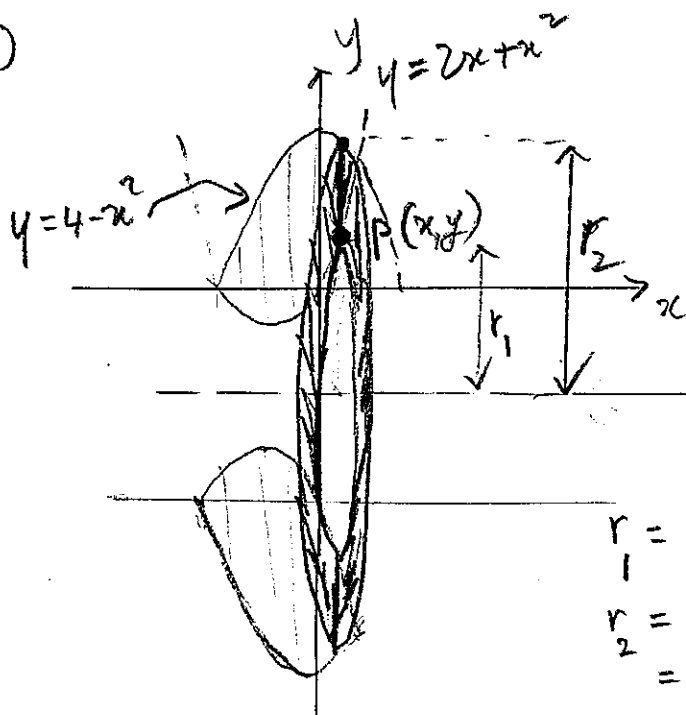
$$= -2\pi \left[x - 5 \ln(x+1) \right]_0^4$$

$$= -2\pi \left[4 - 5 \ln 5 - (0 - 5 \ln 1) \right]$$

$$= 2\pi \left[5 \ln 5 - 4 \right] \text{ units}^3$$

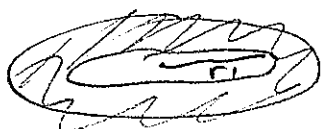


Q7 (b)



Most students did not draw the diagram clear enough to find r_1 & r_2 correctly.

$$\begin{aligned} r_1 &= 2 + 2x + x^2 \\ r_2 &= 2 + 4 - x^2 \\ &= 6 - x^2 \end{aligned}$$



$$dV = \pi (r_2^2 - r_1^2) dx$$

Where $r_2^2 - r_1^2 = (r_2 - r_1)(r_2 + r_1)$

$$\begin{aligned} &= (6 - x^2 - 2 - 2x - x^2)(2 + 2x + x^2 + 6 - x^2) \\ &= (4 - 2x - 2x^2)(8 + 2x) = 4(2 - x - x^2)(4 + x) \end{aligned}$$

$$\therefore V = \lim_{x \rightarrow 0} \int_{-2}^1 4\pi (8 + 2x - 4x - x^2 - 4x^2 - x^3) dx$$

$$= 4\pi \int_{-2}^1 8 - 2x - 5x^2 - x^3 dx$$

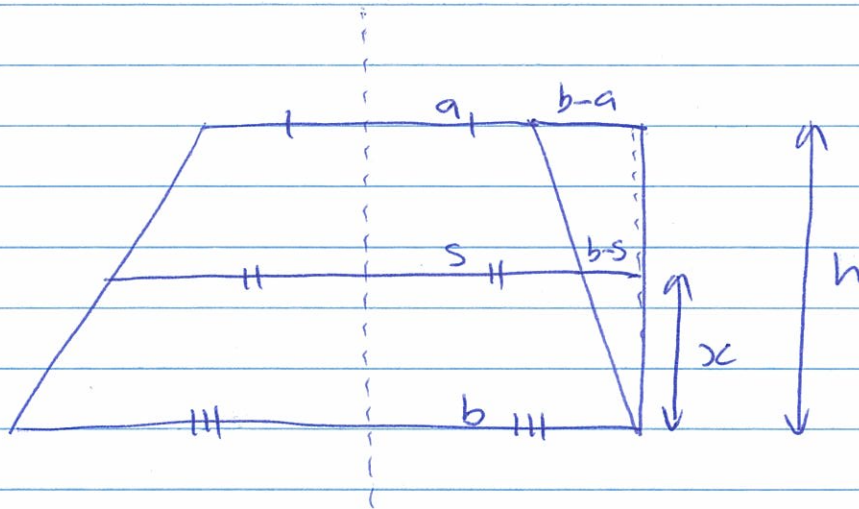
$$= 4\pi \left[8x - x^2 - \frac{5}{3}x^3 - \frac{x^4}{4} \right]_{-2}^1$$

$$= 4\pi \left(8 - 1 - \frac{5}{3} - \frac{1}{4} + 16 + 4 - \frac{40}{3} + 4 \right)$$

$$= 4\pi \left(\frac{63}{4} \right)$$

$$= 63\pi \text{ units}^3.$$

7c.)
(i)



By similar triangles,

$$\frac{b-s}{b-a} = \frac{x}{h}$$

$$b-s = \frac{b-a}{h} x$$

$$s = b - \frac{b-a}{h} x$$

(ii)

$$\delta V = \pi r^2 \delta x$$

For shaded circle,

$$\delta V = \pi s^2 \delta x$$

$$V = \pi \int_a^h s^2 dx$$

$$= \pi \int_0^h \left(b - \frac{b-a}{h} x \right)^2 dx$$

7c.) (i)

$$\begin{aligned} V &= \pi \int_0^h \left(b - \frac{(b-a)}{h} x \right)^2 dx \\ &= \pi \int_0^h \left(b^2 - \frac{2b(b-a)}{h} x + \frac{(b-a)^2}{h^2} x^2 \right) dx \\ &= \pi \left[b^2 x - \frac{b(b-a)}{h} x^2 + \frac{(b-a)^2}{3h^2} x^3 \right]_0^h \\ &= \pi \left\{ b^2 h - \frac{h^2 b(b-a)}{h} + \frac{h^3 (b-a)^2}{3h^2} - 0 \right\} \\ &= \pi \left\{ b^2 h - bh(b-a) + \frac{1}{3} h(b-a)^2 \right\} \\ &= \frac{\pi h}{3} (3ab + b^2 - 2ab + a^2) \\ &= \frac{\pi h}{3} (a^2 + ab + b^2) \end{aligned}$$

8a.) (i) $x^3 + y^3 = -1$

$$3x^2 + 3y^2 \frac{dy}{dx} = 0$$

$$3y^2 \frac{dy}{dx} = -3x^2$$

$$\frac{dy}{dx} = \frac{-x^2}{y^2}$$

(ii) x -intercept, $y = 0$

$$\frac{dy}{dx} = \frac{x^2}{0} = \text{undefined}$$

$$x^3 + 0 = -1$$

$$x = -1$$

tangent at x -intercept has equation $x = -1$

y -intercept, $x = 0$

$$\frac{dy}{dx} = \frac{0}{y^2} = 0$$

$$0 + y^3 = -1$$

$$y = -1$$

tangent at y -intercept has equation $y = -1$.

where curve meets line $y = x$

$$\frac{dy}{dx} = \frac{-x^2}{x^2} \text{ or } = \frac{-y^2}{-y^2} = -1$$

$$x^3 + x^3 = -1$$

$$2x^3 = -1$$

$$x^3 = -\frac{1}{2}$$

$$x^3 = -\frac{1}{2}$$

$$x = -\sqrt[3]{\frac{1}{2}}$$

8a.) (ii) (cont.)

$$\text{at } x = -2^{-\frac{1}{3}},$$

$$x^3 + y^3 = -1$$

$$(-2^{-\frac{1}{3}})^3 + y^3 = -1$$

$$-\frac{1}{2} + y^3 = -1$$

$$y^3 = -\frac{1}{2}$$

$$y = -2^{-\frac{1}{3}}$$

equation of tangent at $(-2^{-\frac{1}{3}}, -2^{-\frac{1}{3}})$

$$y - y_1 = m(x - x_1)$$

$$y + 2^{-\frac{1}{3}} = -(x + 2^{-\frac{1}{3}})$$

$$y = -x - 2 \times 2^{-\frac{1}{3}}$$

$$y = -x - 2^{\frac{2}{3}}$$

8a.) (iii)

$$x^3 + y^3 = -1$$

$$y^3 = -1 - x^3$$

$$= -x^3 \left(1 + \frac{1}{x^3} \right)$$

$$y = -x^3 \sqrt[3]{1 + \frac{1}{x^3}}$$

$$\text{as } x \rightarrow \infty, \frac{1}{x^3} \rightarrow 0$$

$$\therefore y \rightarrow -x$$

8a.) (iv) $\therefore$ oblique asymptote is $y = -x$

